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"Analysis and Optimization of the Position of Stabilising Plates in the TRUST Experiment."

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ABSTRACT

The control of plasma stability is a crucial aspect for the success of fusion reactors. For this purpose, the Tuscia Research University Small tokamak (“TRUST”) project, conceived as an experiment with the dual objective of training specialized engineers and testing new technologies, represents an important challenge in engineering terms for determining the optimal position of the stabilising plates within the machine and ensuring an optimal balance between plasma stability and structural loads.

This thesis proposes a case study focused on optimising the position of the stabilizing plates by analysing two key parameters: the plasma growth rate and the electromagnetic ("EM") loads acting on them. The analysis is conducted using MATLAB and MAXFEA software to simulate and evaluate different configurations, allowing the identification of optimal plate positions that meet plasma stability and structural strength requirements.

1 INTRODUCTION

Research on nuclear fusion represents one of the most promising and ambitious frontiers of contemporary physics. Unlike nuclear fission, which underpins current nuclear power plants and involves splitting heavy atomic nuclei into lighter nuclei, nuclear fusion aims to join light isotopes, such as hydrogen, to form heavier isotopes, releasing huge amounts of energy. This process is what powers the sun and other stars and is seen as a potentially inexhaustible source of clean and safe energy.

1.1 INTRODUCTION TO NUCLEAR FUSION AND EM ANALYSIS IN A TOKAMAK

In this section, we will analyse the various reasons that have prompted nuclear fusion research. Global warming and increased demand for energy have led scientists to look for solutions to obtain energy without emitting large amounts of greenhouse gases and other harmful gases into the atmosphere. These, as we know, in addition to polluting the air we breathe, cause strong variations in climatic conditions, affecting entire biomes and causing economic damage for billions of dollars every year. Currently, only 20% of the energy we produce comes from renewable sources, i.e. with zero impact in terms of greenhouse gas emissions.

Nuclear fusion, like fission, is a zero-emission source of energy. However, unlike nuclear fission, fusion generates much less waste. The slag produced by the smelting is generally encapsulated in concrete envelopes and buried, thus reducing the environmental and economic impact. In addition, with nuclear fusion there is no risk of accidents such as those that have occurred in the past.

In the case of nuclear fission, the process is based on a chain reaction. When a uranium or plutonium nucleus splits, it releases neutrons that can hit other nuclei, continuing the reaction. If this chain reaction is not properly controlled, it can become uncontrollable, leading to explosions or a massive release of radiations. In contrast, there is no chain reaction in nuclear fusion. Melting only takes place under extreme conditions (temperatures of millions of degrees), and if these conditions are not maintained, the process stops. In the event of a problem in a fusion reactor, the reaction shuts down immediately without risk of escalation or explosion.

In addition, the amount of material required to feed a fusion reaction is much lower than that of fission power plants. This makes nuclear fusion a form of energy that requires little fuel and that, in theory, can be considered unlimited

Nuclear fusion occurs when nuclei of light isotopes, such as deuterium and tritium, combine at high temperatures and pressures, overcoming the electrostatic repulsion between protons to form a heavier nucleus (helium), releasing energy in the form of neutrons. The reaction occurs when plasma is obtained.

Plasma can be defined, according to the MIT definition, as "*superheated matter – so hot that the electrons are ripped away from the atoms forming an ionized gas*". To achieve this state, extremely strict conditions are required, the complexity of which depends on the type of confinement method chosen. [5]

1.1.1 Plasma Confinement Methods

In general, plasma confinement methods all seek to achieve the same goal: to provide the proper conditions for plasma formation. There are different types of confinement, including *magnetic, inertial and gravitational*:

- **Inertial confinement** uses very high-energy pulses, often in the form of laser beams or particles, to compress a small amount of fuel until it reaches conditions of sufficient density and temperature to trigger fusion. This method is used, for example, in inertial confinement fusion (ICF) designs, where lasers compress a fuel capsule until it collapses. It is customary to use Deuterium-Tritium pellets as fuel.
- **Gravitational confinement** is the principle on which stars, including our Sun, are based. Here, the force of gravity itself confines and compresses light elements (such as hydrogen) until they melt and release huge amounts of energy. However, replicating stellar-like conditions on Earth is extremely complex and is currently not viable.
- **Magnetic confinement** is the most widely used type of confinement and is carried out using a device called "*tokamak*", a Russian acronym that stands for "*toroidal chamber and magnetic coil*" machine that will be analysed in this thesis. To date, there are no materials capable of withstanding the temperatures that are generated inside the tokamak, (about 100 million ◦C), each material would be ionized and destroyed. To overcome this, a tokamak uses toroidal and poloidal magnetic fields to confine the plasma in a doughnut shape (torus), keeping the charged particles away from the walls of the reactor. Combining these two magnetic fields and the one produced by the plasma itself allows the machine to confine the plasma in the volume of the torus.

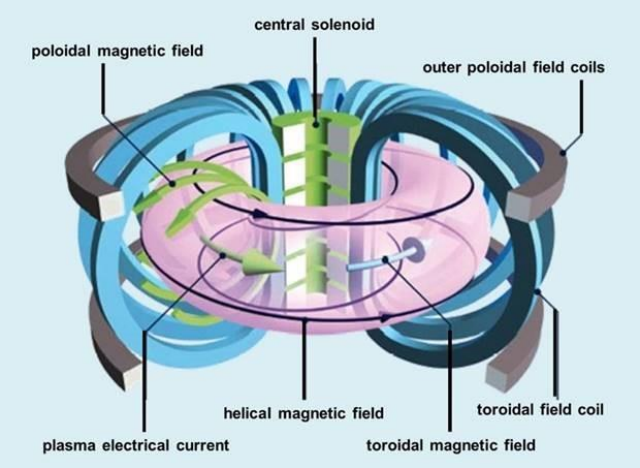


Figure 1: Tokamak plasma overview

The magnetic field is created with various types of coils: "***Poloidal Field Coils*" (PF COILS)** that generate a poloidal magnetic field, and the **"*Toroidal Field Coils*" (TF COILS)** that allow you to generate a toroidal magnetic field. By combining these two magnetic fields, and the same magnetic field generated by the plasma current, we obtain a helical field; all this allows the plasma to be trapped. In this context, stabilising plates, metal structures placed at specific points in or around the tokamak chamber, are of particular interest for this thesis. Their main function is to absorb and dissipate the currents that develop because of magnetic instabilities. When the plasma tends to move, these plates generate induced currents that create a magnetic field of reaction, helping to "dampen" unwanted movements of the plasma and keep it stable. These plates can be made of various materials and in the TRUST project they are made of copper.

In the TRUST project, the reference machine that we will analyse, the goal is to test high-temperature superconducting materials (HTS), intended for the production of coils and magnets necessary for the operation of the tokamak.

There is also a central solenoid, consisting of several modules, which is used to start, sustain and control the plasma's switching on and off precisely.

Electromagnetic (EM) analysis is critical to understanding and optimizing the behaviour of the plasma within the tokamak, ensuring stability and efficiency in the melting process. In a fusion reactor, such as the tokamak, magnetic fields confine the plasma to high temperatures, allowing fusion reactions. However, these fields also generate intense forces that act on the reactor structures, and understanding their behaviour is crucial for the safety and performance of the system.

The aim of this thesis is to verify the most critical conditions, in particular regarding the "*disruption forces*", which occur when the plasma suffers a loss of confinement. These forces can be extremely damaging to the internal structures of the reactor. In this context, one of the main objectives is to identify the optimal position of the stabilising plates in the reference machine, ensuring that it can withstand the electromagnetic forces generated during disruptions.

To conduct these analyses, we use MAXFEA software, a *Finite Element Method* (FEM) solver, which allows us to model the behaviour of the plasma inside the vacuum chamber. MAXFEA allows you to simulate various operating situations, analysing in detail how magnetic fields interact with the structural components of the reactor. This gives us the ability to determine, as in our case, the optimal position of the stabilising plates inside the tokamak, ensuring that they can effectively reduce the effects of electromagnetic forces and ensure the overall stability of the system.

Thanks to these simulations, we can predict and manage the risks related to electromagnetic forces, optimizing not only the efficiency of plasma confinement, but also the structural safety of the machine itself. The use of advanced tools such as MAXFEA allows us to explore the worst case of "disruption forces". The aim of the thesis is therefore to identify the ideal position for the stabilising plates, ensuring the best possible balance between the "*Growth Rate*" (GR) value and values referring to the forces that are generated inside the tokamak.

1.2 MACHINE REFERENCE: TRUST

TRUST (Tuscia Research University Small tokamak) is an experimental facility designed to study and improve plasma confinement and develop advanced technologies related to nuclear fusion. This research platform represents a key step towards the realization of operational fusion reactors, addressing the technical challenges that currently exist.

Among the many objectives of TRUST, the main ones are:

1) Train the next generation of engineers in the field of nuclear fusion: The facility offers a unique opportunity for young engineers to immerse themselves in an advanced research environment, where they can acquire the practical and theoretical skills needed to tackle the complex technical problems related to fusion. TRUST aims to train experts capable of designing and operating the reactors of the future.
2) Testing innovative materials: One of the main aims of the project is to test and develop new materials that can withstand extreme conditions inside the tokamak, such as ultra-high temperatures and electromagnetic forces. In particular, TRUST allows for the experimentation of materials for the first wall of the reactor, which must be able to withstand intense heat and direct interaction with the plasma.

3) Improving plasma confinement technologies: The facility allows for the study of new magnetic configurations and confinement techniques to ensure plasma stability and increase the efficiency of the melting process. TRUST thus provides an essential platform for the development of advanced technological solutions that can be applied to future commercial reactor designs.
4) Simulate extreme conditions: TRUST's infrastructure enables large-scale experiments to be carried out in which the behaviour of plasma and support structures under extreme operating conditions are analysed. This allows researchers to optimize the design of fusion reactors and verify the ability of the structures to withstand the critical conditions that arise during the fusion process.
5) One of the standout features of TRUST is its symmetrical poloidal coil system, which is designed to provide remarkable versatility in plasma configuration. Thanks to this peculiarity, TRUST is able to operate with different magnetic configurations, adapting to the needs of experimentation and analysis. In particular, the device is optimized to work with three main plasma configurations:

    - Downward Single-Null (DSN), characterized by a null point at the bottom of the vacuum chamber, ideal for stability studies and plasma control in asymmetrical configurations.
    - Upper Single-Null (USN), which has the null point at the top, allowing comparative analysis with the DSN configuration.
    - Double Null (DN), with null points both above and below, particularly suitable for analyses that require magnetic symmetry and reduction of thermal stresses on the device wall.

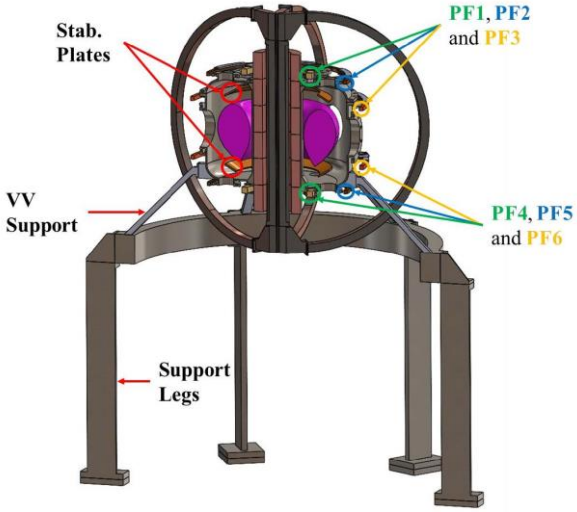


Figure 2: TRUST Structure

1.2.1 Insight into Growth rate, single null and X-point

The "Growth Rate" within a tokamak is a parameter that measures the rate at which instabilities develop in the plasma. These instabilities are unintended changes in the shape or density of the plasma that can lead to a degradation of the confinement or even the total loss of the plasma, compromising the operation of the reactor. [9]

In the theory of magnetohydrodynamic instabilities (MHD), the "Growth Rate" quantifies the rate at which a perturbation grows over time. In practice, it indicates how quickly a small deviation from the stable configuration is amplified, causing the plasma to go out of balance. [9]

A high growth rate means that the plasma is highly unstable, and these instabilities could expand so rapidly that they damage the plasma's confinement. The goal of the control is to keep the growth rate low, so that the plasma remains stable and confined for longer, improving the efficiency of the reactor.

The configuration of the plasma significantly affects the properties of the reactor and the behaviour of magnetic confinement. In the experimental TRUST reactor, one of the possible configurations is the "Single Null Diverted" (SND), i.e. a configuration with only one null point of the magnetic field inside the torus. This null point is known as the X-point, and it is a crucial element in plasma control and stability. [8] [9]

The term "Single Null" refers to a plasma configuration in which there is a single point, the X-point, where the magnetic field cancels out. In this configuration, the plasma is magnetically confined, but near the X-point, the magnetic field lines open to an area called the divertor. The divertor is a region designed to manage the flow of particles and heat that comes out of the main plasma, preventing them from touching the walls of the reactor and damaging the structure.

The X-point is fundamental because it represents the point at which the magnetic field cancels out and the field lines separate: one part of the field lines borders the plasma, while the other part connects the plasma to the divertor. This separation is crucial to reduce erosion of the reactor's inner walls and improve particle and heat flow control.

The "Single Null" configuration with an X-point offers some advantages over other configurations. Firstly, it provides good plasma control and allows you to effectively manage impurities and heat, directing them towards the divertor. It also simplifies reactor engineering and design, as there is a single critical point to monitor and optimize. However, compared to more advanced configurations such as "Double Null Diverted" (DND), which uses two X-points, "Single Null" can be less efficient in heat management in large reactors. [8] [9]

1.3 Presentation of the Problem: Objective of the Thesis

This thesis aims to address one of the critical issues in the operation of tokamaks: the role of stabilising plates in ensuring plasma stability. Stabilising plates are critical components for controlling magnetohydrodynamic (MHD) instabilities, which can compromise plasma confinement, with significant impacts on reactor efficiency and safety. They are designed to mitigate MHD instabilities, such as resistive modes and ballooning instabilities, which can lead to confinement losses and damage reactor structures. Through accurate control of currents and magnetic fields in stabilising plates, it is possible to mitigate these instabilities, improving the quality of plasma confinement and moving closer to the goal of sustainable and reliable nuclear fusion. [6] [8]

The main objective of this thesis is to analyse the behaviour of stabilising plates in different operational scenarios, evaluating their performance and identifying possible improvements. This study aims to optimize the stability of plasma within the TRUST tokamak, thus helping to develop innovative solutions to address the technical challenges associated with plasma confinement and control. The scenario considered for the analysis was the END Of Flat-top (EOF) scenario, for the DSN configuration; i.e. the moment when, due to the achievement of parameters such as: Current, Pressure, Poloidal Magnetic Field generated by the plasma itself, it is necessary to turn off the plasma. This operation is due to the achievement of the technological limit of the Central Solenoid and all the coils, which work together as a primary circuit that induce current by mutual induction in the plasma, consequently acting as a transformer. Given the order of magnitude of the currents flowing ($1 * 10^6$) A, it is not possible to keep them active for too long at a time, due to the Joule Effect which leads to very high temperatures, compromising the structural stability of the tokamak. In this

shutdown phase, where we bring the plasma current I_p to 0 in order to ideally shut down the structure safely, it happens, as we will analyse later, that instead of a safe shut-down, various unpredictable physical phenomena can occur, which lead to conditions called "off-normal" better known in the technical field as "disruptions". [6] [7] [8] [9]

2 OBTAINING CENTROID COORDINATES OF STABILISING PLATES AND EM ANALYSIS

2.1 INTRODUCTION TO THE APPROACH USED

For the production and study of stabilising plates within the TRUST project, two models have been applied, the Filamentary model, where only the vertical movement of the plasma is taken into account, and the "Radial and Vertical Plasma Position Method" (RZIP) model, where vertical and radial movements are taken into account, together with the total variations of the current I_p of the plasma, these two models made it possible to obtain a series of possible centroid coordinates (i.e. point corresponding to the geometric center of the plate). The analyses were carried out at the EOF. The Filamentary model and RZIP model analyses are 0D analyses, where we evaluate the "Growth Rate" ("GR"), i.e. the rate at which a plasma instability increases over time. The GR is a key parameter for assessing the magnetohydrodynamic stability (MHD) of the plasma, a high GR indicates that the instability develops rapidly, making it more difficult to control the plasma. [1] [2] Downstream of the analyses using the Filamentary model and the RZIP model, it was decided to also carry out analyses concerning the loads (Electromagnetic Forces) applied on the plates using MAXFEA software, thus evaluating the 2D case. These analyses were carried out, as well as for the analyses using the Filament and RZIP models, in the EOF for two scenarios: which are a Downward "Vertical Displacement Event" (VDE) in the FAST case and in the SLOW case. In these configurations, the plasma follows a downward trend and collapses on the lower first-wall, potentially damaging the structure due to loads and temperatures, which can reach extremely high values. The difference between FAST and SLOW cases is in the rate at which the plasma loses electro-magnetic energy, depending on the time for the current to decay to zero. [7]

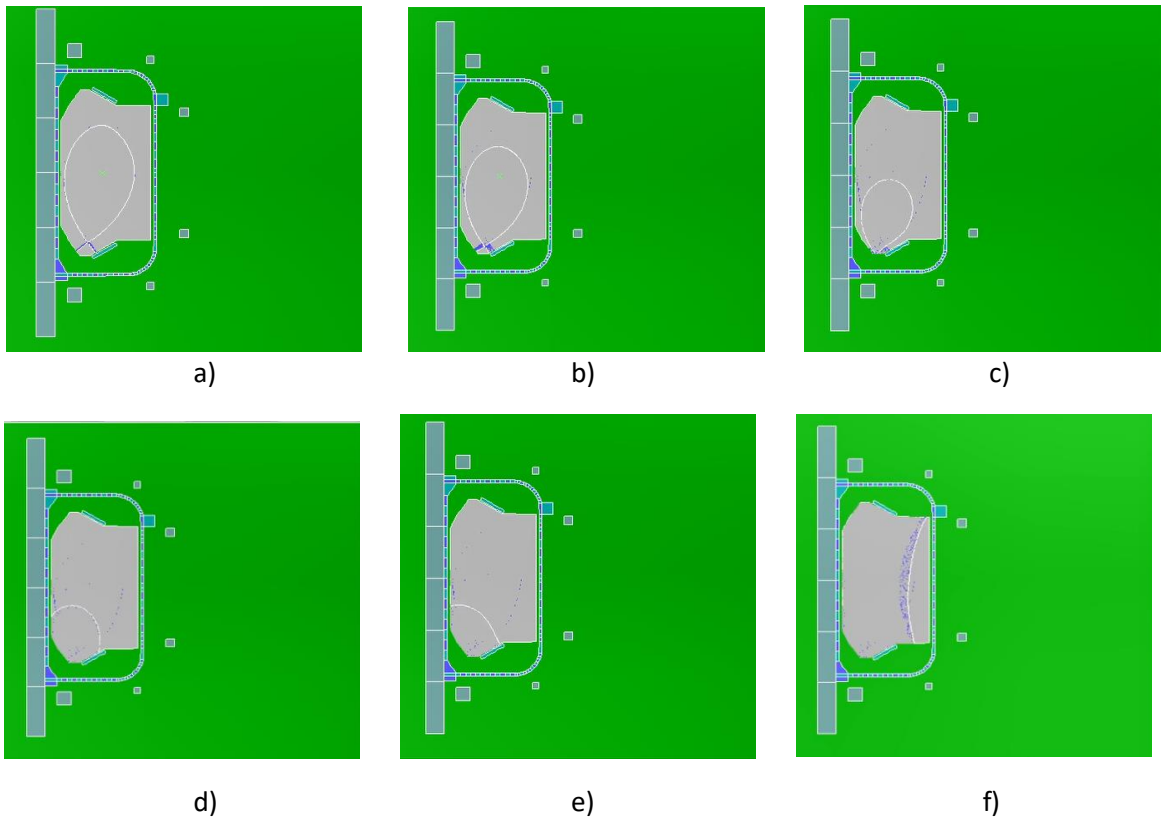


Figure 3: a,b,c,d,e,f show the progress of the disruption in various snapshots with the MAXFEA software

2.1.1 Filamentary Model

The Filamentary Model is a technique used in the analysis of magnetic fields and in the modelling of induced currents in devices such as tokamaks. This technique is particularly useful for simulating the interaction of plasma with coils and surrounding metal structures, to predict the electromagnetic forces and fields generated. In the most simplified model of unstable vertical motion, the plasma ring is approximated by a rigid filament with current, which can only move in a vertical direction. [2]

The vertical force exerted on this filament by a radial magnetic field can be expressed by the following formula [2]:

$$m_p \ddot{z}_p = -2\pi r_p I_p B_r\left(z_p, I_a, I_e\right) \quad (1)$$

Where m_p is the mass of the plasma and B_r is the radial magnetic field with component (r_p, z_p), which depends on the position z_p of the vertical filament and the currents in the passive and active circuits, respectively I_a, I_e . [2]

The fundamental principle behind the Filamentary Model is to divide the electric current (in a coil, or in the plasma itself) into a series of "filaments" or linear conductors, each with a defined current. This discretization allows you to transform a continuous problem into a discrete one, facilitating numerical calculations.

In addition, the Filamentary Model allows you to simulate induced currents on conductive surfaces. This is useful for understanding how the magnetic field of plasma can induce eddy currents, which could in turn affect the stability and control of the plasma.

The method is based on Green's functions for the magnetostatic, Green's functions are used to describe the contribution of the magnetic field generated by a single filament-like current element. In the method, the total magnetic field is obtained by integrating the contribution of each filament, thanks to the linearity of Green's functions. [11]

2.1.2 Model RZIP

A more specific and quantitative analysis is offered by the RZIP model, which is the reference model used in the TRUST project. RZIP is a simplified and more precise model for the analysis of stability and magnetic control in tokamaks. It represents a reduced, linear version of the overall system, particularly useful for plasma stability simulations and for the control of eddy currents and magnetic perturbations. This model is based on the calculation of the average of the forces acting on the plasma current under consideration. [2]

RZIP is based on a linearized model, which allows to analyse the dynamic response of plasma with respect to small disturbances, without having to simulate the entire nonlinear behaviour.

The model reduces the complexity of the complete geometry of a tokamak, focusing on the key aspects of magnetic response and plasma position.

The analyses and simulations made with the RZIP model are based on the following physical laws that describe respectively the dynamics of the circuits, the equilibrium of the vertical and radial forces, and the evolution of the plasma current [2]:

$$L_c\dot{I}_c + R_cI_c + \dot{\psi}_{cp} = Vc \quad (2)$$

$$2\pi r_p I_p\{B_r^{ext}(r_p, z_p, I_c)\} = 0 \quad (3)$$

$$2\pi r_p I_p\{B_r^{ext}(r_p, z_p, I_c)\} + F_r(r_p, I_p) = 0 \quad (4)$$

$$\frac{d}{dt}[L_p(r_p)I_p] + R_pI_p + \dot{\psi}_{cp} = 0. \quad (5)$$

Where $I_c = [I_aI_e]^T$ it encompasses both active and passive currents, ψ_{cp} is the contribution flow that is transferred from the plasma to the active and passive circuits, $\langle B_r^{ext} \rangle$ and $\langle B_z^{ext} \rangle$ are the radial and vertical field components generated by the average of the external circuits along the distribution current (in practice an estimate of this distribution could be obtained in real time through a magnetic code of equilibrium reconstruction). [2]

The RZIP plasma model (R, z, Ip) is based on the linearization hypothesis, i.e. that small changes in PF coil voltages lead to small changes in plasma current, PF currents, vacuum vessel currents, and radial plasma.

"These small variations occur around a certain state of unperturbed equilibrium. It is assumed that the equilibrium parameter (equilibrium profile) Λ ≡ βp + li/2 remains fixed, although variations in this parameter may be introduced as a disturbance." [12]

2.1.3 Methodology applied to Filament & RZIP models

The analysis of the two models was carried out using MATLAB software. In particular, the work carried out consists in defining an area within the "*vessel*" in which to test the different configurations of the centroids of the stabilising plates, delimiting the points to be considered.

This process allows the two models to be applied to calculate the behaviour of the plasma growth *rate* at different points in the area of interest.

The developed script generated a grid of points, each represented by a pair of vertical and horizontal coordinates. For each of these points, the "*Growth Rate*" (GR) was calculated using the two models, with the aim of identifying the most stable areas where the GR is minimal, in order to optimize the position of the stabilising plates. Limits have been set to ensure that the plates do not exceed the geometry of the reactor ("*first wall*") and that, given the size of the plate, the inner side is at least 2 mm away from the *first wall*. Another important aspect concerns the distance between the centroids in the grid used for the analysis. To ensure accurate discretization of the domain and effectively capture changes in the growth *rate*, the mesh was designed with a centroid spacing of approximately 0.60 cm along the horizontal axis and 0.55 cm along the vertical axis. An inclination of -29.7° has been imposed on the plates for each pair of plates, to ensure their parallelism with respect to the first-wall. The thickness and length of the plates was also assumed to be constant for each pair of plates, with a value of 0.1 cm and 0.9 cm.

These sizes were chosen based on a trade-off between the desired resolution and computational efficiency. A smaller distance between the centroids would have increased the number of grid elements, increasing the detail but also the computational time required for simulations. Conversely, longer distances could have led to a loss of precision in the results. The choice of 0.60 cm and 0.55 cm therefore ensures optimal balance.

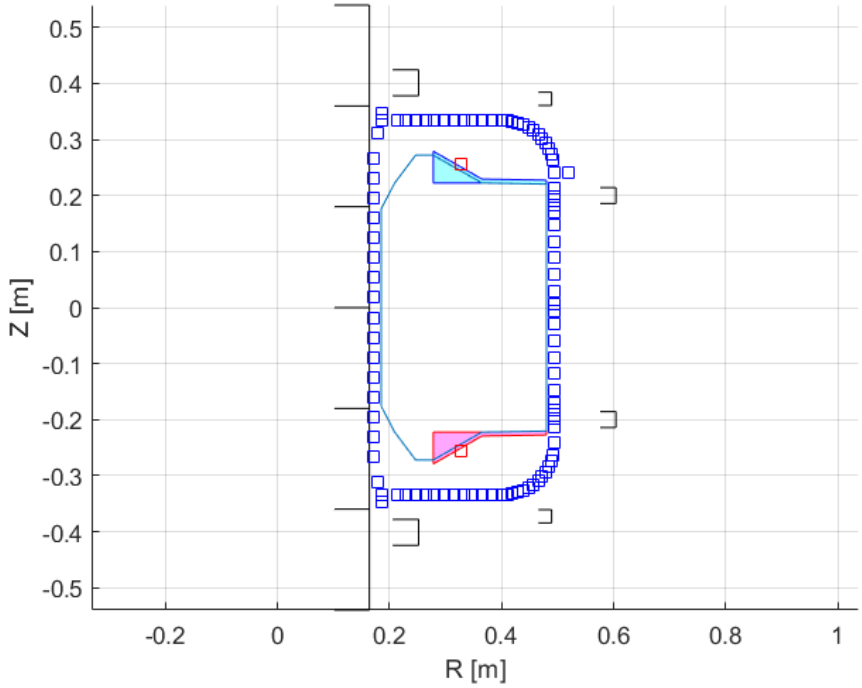


Figure 4: Plot of the positioning of the centroid of the V_0 version (Reference), in blue and pink the areas in the tokamak to be excluded from the possible combinations, ensuring a minimum distance of the plate from the first wall (light blue line)

The possible positions of the centroids, as a result of the geometric limits applied, have increased from 400 to 360.

2.1.4 MAXFEA Analysis and methodology

The second part of the thesis concerns the analysis of the loads applied on the stabilising plates, due to the Lorentz Force. This analysis was possible thanks to the use of the EM analysis software, MAXFEA, a tool written by Pietro Barabaschi [13]. MAXFEA is a 2D FEM (Finite Element Method) solver with triangular geometric elements that is able to solve the Grad-Shafronov equation. [4]

MAXFEA allows the dynamic evolution of plasma under normal or non-normal conditions (e.g., "disruptions"). However, MAXFEA has a limitation, namely that it can only solve an axisymmetric geometry.

The analyses carried out with the MAXFEA software, as already explained, are analyses based on the EOF scenario, with MAXFEA more specifically we are going to analyse the electro-magnetic energy. In the simulations carried out, we analysed an event called "Vertical Displacement Event" (VDE) in the downward condition. [7]

These VDEs follow phases:

- Growth of instability: Unpredictable problems are highlighted, causing the plasma to lose its nominal (ideal) position. The instability consequently leads the plasma to move up or down the chamber, with a certain speed that can be controlled through the GR (Growth Rate) parameter, a parameter that, as explained above, helps us to have a good estimate of how much speed the plasma moves up or down. As a result, the vertical balance of forces is no

longer respected due to the radial field variation. The plasma, leaning on the structures, passes from the diverter configuration to the limiter configuration. [8]

- The Safety Factor (q), a crucial parameter in plasma stability analysis, defines the magnetic configuration required to keep the plasma safely confined within the tokamak. This parameter represents the ratio between the number of toroidal and poloidal rotations that a magnetic field line makes before closing in on itself. [9] [14]

  A threshold value of q of 2 is commonly considered the lower limit to ensure stable operating conditions. When the q value falls below this threshold, the plasma enters a critical condition in which magnetohydrodynamic stability (MHD) is compromised. [9] [14]

  When the limit value of the Safety Factor is reached, the thermal quench phase is triggered, characterized by a drastic loss of thermal energy inside the plasma. During this phase, the rapid drop in temperature leads to a decrease in internal pressure and an accentuation of the currents induced in the walls of the tokamak, with consequent structural stresses that can damage the apparatus. [8] [9] [10]

- "Thermal quench": When the Safety Factor (q) falls below the critical value of 2, the plasma enters a Thermal Quench (TQ) phase, characterized by a rapid loss of thermal energy. During this phase, there is a dramatic drop in plasma temperature, accompanied by a decrease in internal pressure. [9]

In the simulations carried out with MAXFEA, the Thermal Quench phase is not directly analysed in terms of plasma temperature or density, but is represented through the parameter β_{pol} (Poloidal) [6], a parameter that corresponds to the ratio $\frac{2\mu_0\langle p\rangle}{B_\theta^2}$ (6), which expresses the ratio between the average plasma pressure ⟨p⟩ and the poloidal magnetic field B_θ^2 squared. During the Thermal Quench, β_{pol} it undergoes a rapid decrease to values close to zero, thus simulating the energy-thermal collapse of the plasma. At the end of the Thermal Quench, the plasma moves on to the next phase, known as Current Quench, in which the plasma current progressively decays, with decay times varying depending on the system configuration. [6] [7] [8] [10]

- "Current quench": After the Thermal Quench phase, the plasma undergoes a rapid loss of electromagnetic energy, entering the Current Quench (CQ) phase. This phase is characterized by the decay of the toroidal plasma current, which occurs due to the collapse of the magnetic configuration of the plasma. The loss of current is accompanied by a descending movement of the plasma towards the containment chamber (first wall), a phenomenon known as vertical displacement.

Current decay can occur with two main dynamics:

- FAST Current Quench: a rapid decay of the current, typically on the order of milliseconds
- SLOW Current Quench: a slower decay, with an extended time scale.

The decay time, which separates the FAST and SLOW behaviours, is mainly determined by the electrical resistance of the plasma and the geometry of the confinement system. This step is crucial for the design of the tokamak's structural components, as high eddy currents can cause significant damage to the walls and conductive elements. Accurate modelling of QC, including its effects on structures, is essential to ensure operational safety and prevent catastrophic failures. [7] [10]

A FAST current quench results in a higher derivative of the poloidal flow with respect to time, which according to Faraday's law implies a higher induced voltage in the structures, which translates into a higher induced current in the structures.: $\frac{\partial \psi}{\partial t} = -f_{em}$ (7)

With a SLOW "Current quench", there is a larger electro-magnetic load distributed over time, consequently the derivatives of the poloidal flux with respect to time are lower and therefore there is a lower induced voltage that generates reduced eddy currents. From a thermal point of view, a SLOW situation is negative due to the greater time spent by the plasma on the structures, as it shuts down.

2.1.4.1 Methodology and hypotheses

The work carried out until now was to focus on 8 pairs of coordinators, which were the most plausible for the real construction and obtaining the exact coordinates of the vertices of the plates. This result was possible thanks to the writing of a special script, which used as data the geometries of the plates, equal to a length (long side) of 9 cm and a width (short side) of 1 cm and the inclination with respect to the horizontal plane of the plate of about -29.7°. This angle is necessary to respect the parallelism of the plates with respect to the "first wall" and allow the construction hypotheses. Once the coordinates of the centroids and vertices of the various possible positions of the plates were obtained, the next step was to use a mesher, i.e. an algorithm that allows you to create a mesh starting from a defined geometry. The type of Mesh obtained is based on the Finite Element Method (FEM), i.e.: "a subdivision of a continuous domain into smaller elements, such as triangles or quadrilaterals, to facilitate the numerical solution of differential equations" [3], a necessary condition for the correct and effective subsequent interpretation on MAXFEA. Next, I calculated the vertices of the plates, which are rectangular in shape, and defined the appropriate angle for each plate. Once the vertices of the stabilising plates were determined, I generated a mesh for each of these configurations, this was essential for the second part of the thesis analysis, i.e. the analyses concerning the loads applied to the stabilising plates.

The density of the meshes is one of the elements that can allow you to obtain a higher quality of results, as already explained above, the denser the mesh, the more precise and reliable results can be obtained. At this point, it was necessary to impose a mesh size, which depending on the coordinate under analysis, can vary from 28,000 elements to 30,000, this was necessary given the nature of the MAXFEA analysis software; in fact, it is precisely with these values that the MAXFEA software manages to be as efficient and precise as possible.

Once the meshes were obtained, the next steps were to transform them into a format readable by MAXFEA, for this purpose a Fortran script was used.

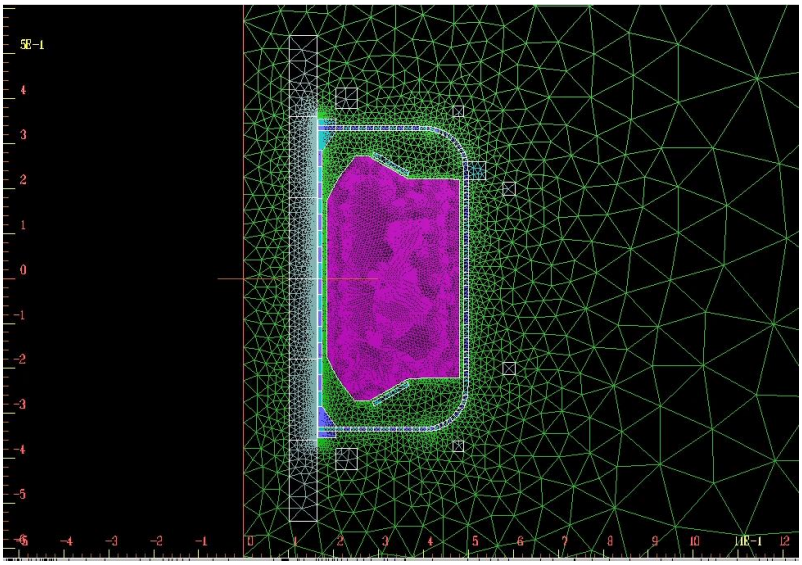


Figure 5: What the tokamak looks like in MAXFEA after loading the mesh, the first wall, the plasma region (pink), the coils, and the rectangular-shaped stabilising plates

Once the mesh was loaded into MAXFEA, it was necessary to define the various materials to be assigned to each element of the tokamak. In particular:

- All passive structures have been considered as AISI 316-LN steel
- The areas affected by the PORTs, i.e. the opening areas in the vessel, dedicated to the insertion of diagnostic elements, have been adequately modelled in terms of equivalent resistance
- All coils for this analysis are considered High-Temperature Superconductors (HTS) which are materials that exhibit superconducting properties at significantly higher temperatures than traditional superconductors [15]
- The stabilising plates are made of copper
- The next step is to define the dummy circuits on which MAXFEA relies to solve the PLASMA Equilibrium.

From the MAXFEA simulations it was possible to save the curves between [1-946], this was necessary to make sure we were able to save such information:

- Iplasma: Parameter that indicates the trend of the plasma current, which in the "Current Quench" phase, goes to 0 over time
- Ipol@af0: Parameter defining the "halo Current"
- Fz(PltFsup/PltFbot): That is the loads applied to the stabilising plates

Halo Currents are currents that are generated during the Current Quench phase, when the plasma loses its electromagnetic energy and moves towards the lower wall of the tokamak. This phenomenon occurs in the last moments of the plasma's life, when it opens onto the structure. This opening creates a limited region on the plasma surface where currents are established between the plasma and the surrounding structures, mainly in a poloidal direction. These currents, known as Halo Currents, close through the structure of the tokamak. [8] [9] [10]

The necessary condition for the generation of Halo Currents is that the plasma is located very close to the walls and has a resistivity comparable to that of the surrounding conductive structures. This situation typically

occurs during the Thermal Quench and Current Quench phases, when the plasma loses thermal and electromagnetic energy. [8] [9] [10]

Measuring Halo Currents is critical because they represent local loads that can vary significantly depending on where the plasma releases energy to the structure.

Unlike Halo Currents, Eddy Currents are induced in conducting materials exposed to time-varying magnetic fields, as described by Faraday's Law. These currents occur within the structure itself and are caused by Faraday's law, according to which a changing magnetic field generates an electromotive force (EMF) in the conductor.

Eddy currents are global loads and are in a toroidal direction, as opposed to Halo currents which are in a poloidal direction. Consequently, theoretically, the "Eddy Currents" to generate Lorentz Forces

$$\bar{F} = \bar{J} \times \bar{B} \quad (8)$$

they act with the POLOIDAL Magnetic Field, while the "Halo Currents" being in the poloidal direction to generate Lorentz Forces act with the Toroidal magnetic field. [1] [7] [10]

As we will see later, it should be noted that in this thesis it was possible to analyse the loads generated by the Eddy Currents, on the contrary it was not possible to quantitatively analyse the applied loads generated by the Halo Currents. This topic will be explored and considered later.

3 RESULTS

3.1.1 Filamentary Model Analysis

The results of the analysis of the filamentary model have shown a clear GR trend, what we can see from the plot below, is a decreasing GR trend approaching the first-wall, and increasing as it moves away. This implies, as expected, greater stability near the first-wall.

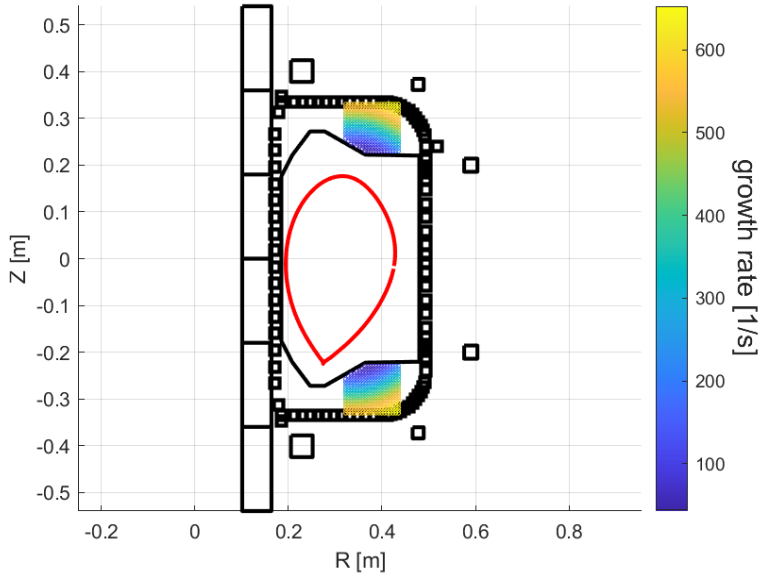


Figure 6: Qualitative trend of the GR by filamentary model inside the TRUST tokamak

To perform a more detailed analysis, 8 pairs of centroid coordinates with a Z coordinate close to the reference value (V_0) were selected. The results obtained show that it was possible to identify centroid configurations that increase the value of the Growth Rate (GR) by up to 61% with respect to the reference value (V_0) of the plates. The trend in the results shows, in general, an increase in the Growth Rate (GR) as the distance from the first-wall increases. However, a small variation in centroid coordinates from the reference configuration can lead to improvements in the GR. However, these improvements, being of a small percentage, do not justify moving the plates away from the reference position V_0.

Table 1: Results obtained by the Filamentary model where a negative change value indicates an improvement in the growth rate compared to the reference value.

Versions analysed	R (Top) [m]	Z (Top) [m]	R (Bottom) [m]	Z (Bottom) [m]	Growth Rate [s^{-1}]	Variation [%]
V0	0,3262	0,2559	0,3262	-0,2559	118,1019	0
V1	0,3288	0,2526	0,3288	-0,2526	100,5969	-14,8218
V2	0,3406	0,2526	0,3406	-0,2526	100,2983	-15,0747
V3	0,3524	0,2526	0,3524	-0,2526	104,9079	-11,1716
V4	0,3642	0,2526	0,3642	-0,2526	115,0409	-2,59178
V5	0,3760	0,2526	0,3760	-0,2526	131,8279	11,62217
V6	0,3878	0,2526	0,3878	-0,2526	156,6046	32,60123
V7	0,3996	0,2526	0,3996	-0,2526	190,2952	61,12797

The filamentary model represents an extreme simplification of the behaviour of plasma, in which it is approximated as a rigid filament that carries current and moves exclusively in the vertical direction. Although useful for preliminary analysis, the model has significant limitations in terms of accuracy.

The filamentary model does not consider the complete dynamics of the plasma, such as current distribution or radial and poloidal movements. As a result, the results obtained are only useful for qualitative estimates. For example, the **Growth Rate** (GR) calculated with this model is significantly lower than most advanced models, showing that it captures only the order of magnitude of the phenomenon without providing reliable quantitative values.

Despite these limitations, the filamentary model remains a useful tool for qualitatively understanding the behaviour of plasma during vertical instabilities. Due to its simplicity, it is ideal for preliminary analysis or to obtain an overview of the dynamics of the system.

Comparing the filamentary model with more precise approaches, such as the RZIp model, its approximate nature clearly emerges. The RZIp model, which considers both radial and vertical movements and integrates the plasma current distribution, provides estimates of the GR much closer to reality. This confirms that the filamentary model is mainly suitable as a preliminary tool, while more detailed analyses require more complex models.

3.1.2 Analysis of RZIp model results

The RZIp model represents a significant improvement over the filamentary model in the description of vertical instabilities in tokamak plasmas. It integrates both radial and vertical movements of the plasma, also considering the distribution of the total current (I_p) and provides quantitatively more accurate results for the calculation of the Growth Rate (GR) and for the analysis of the magnetohydrodynamic stability (MHD). The comparison between the filamentary model and the RZIp model confirms the trend of the GR within TRUST, confirming the tendency to increase, as the distance from the First-wall increases.

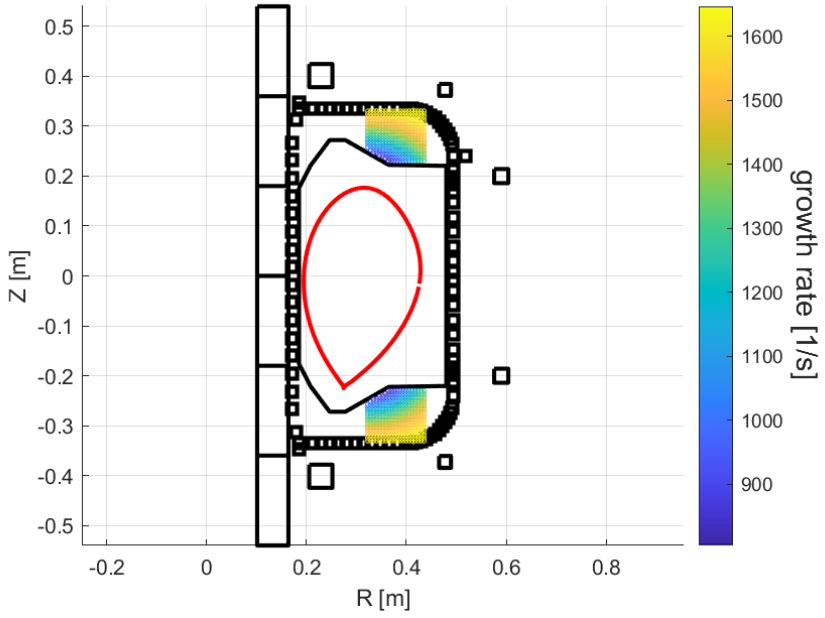


Figure 7: Qualitative and quantitative trend of the GR by RZIp model within the TRUST tokamak

Similarly to what was done for the filamentary model, the same 8 pairs of centroid coordinates were selected in the RZIp model, in order to allow a direct comparison between the two approaches. From the results of the analysis, it emerged that, even in the RZIp model, it is possible to identify centroid configurations capable of increase the value of the Growth Rate (GR) by up to 23% with respect to the reference value (V_0) of the plates. Similar to the Filamentary model, it is possible to observe even more clearly a trend in the Growth Rate (GR) which tends to increase away from the first-wall. As in the Filamentary Method, it is also possible to obtain a better GR value by varying the coordinates with respect to the reference position. However, considering that the percentage changes are even lower, these improvements are so small that there is no justification for moving the plate from the reference configuration V_0

Table 2: Results obtained by RZIp model where a negative change value indicates an improvement in the growth rate compared to the reference value

Versions analysed	R (Top) [m]	Z (Top) [m]	R (Bottom) [m]	Z (Bottom) [m]	Growth Rate [s^{-1}]	Variation [%]
V0	0,3262	0,2559	0,3262	-0,2559	987,7471	0
V1	0,3288	0,2526	0,3288	-0,2526	951,2439	-3,6955
V2	0,3406	0,2526	0,3406	-0,2526	974,5481	-1,3362
V3	0,3524	0,2526	0,3524	-0,2526	1008,5129	2,1023
V4	0,3642	0,2526	0,3642	-0,2526	1051,7509	6,4797
V5	0,3760	0,2526	0,3760	-0,2526	1102,6727	11,6351
V6	0,3878	0,2526	0,3878	-0,2526	1159,5612	17,3944
V7	0,3996	0,2526	0,3996	-0,2526	1220,6465	23,5788

Unlike the filamentary model, the result obtained with the RZIp model is much closer to physical reality, providing a more reliable quantitative estimate. This further highlights the predominantly qualitative nature of the filamentary model, while the RZIp model remains a more accurate tool for plasma stability analysis and identification of optimal configurations. [2]

Although more complex than the filamentary model, the RZIp remains simple enough to be used in preliminary analysis and for the design of control systems.

A comparative analysis of the Growth Rates obtained from the Filamentary and RZIp models highlights some significant trends. In general, the values of the Growth Rates are higher when the stabilizer plates are positioned further away from the first wall of the device. This behaviour is expected, as greater spacing from the plates reduces the effectiveness of magnetic control over instabilities.

For the Filamentary model, the Growth Rates are generally of the order of 10^2, while for the RZIp model they are higher values that fluctuate in the order of 10^2 and 10^3 depending on the geometric configuration and the positioning of the plates. This confirms that while the two models have differences in calculation details and basic assumptions, the overall trends remain consistent.

3.1.3 MAXFEA Results Analysis

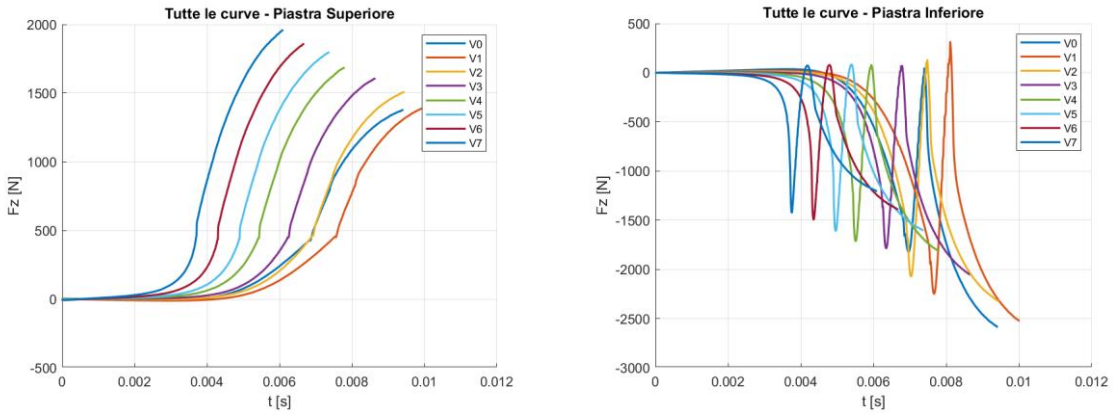
Following the analyses carried out with the MAXFEA software, in which various disruptions were simulated, it was possible to derive a series of curves representative of the associated electromagnetic dynamics. These curves describe the evolution of plasma currents, Halo Currents, and loads applied to the stabilising plates. The data was exported using a dedicated script developed in MATLAB, which allowed the processing and representation of curves in the form of graphs. These graphs provide a detailed view of the time trend of the analysed parameters, highlighting their variations during the different phases of disruption. A characterizing element regarding the choice of the optimal position for the stabilizing plates, in addition to being as already widely specified in the previous chapters the GR, is certainly the load that our stabilising plates receive. To allow a correct comparison between the results obtained by means of the RZIP and the Filamentary models, the same 8 pairs of coordinates previously analysed in detail have been analysed by MAXFEA. The study was carried out for the VDE Downward configuration, in the FAST and SLOW cases.

3.1.3.1 Analysis Results MAXFEA EOF VDE-Downward Fast Configuration

By first analysing the FAST configuration, but similar results are obtained in both configurations, it is possible to visualize the trend of the loads applied on the stabilising plates in the various possible chosen positions of the plates.

The first thing that can be noticed, taking into consideration the curves on the Lower Plate, is that electromagnetic loads on the lower stabilising plate are significantly higher than those on the upper plate. This phenomenon is related to the fact that, being a (VDE) Downward, the plasma, during its dissolution, gets closer to the bottom plate.

From the graphs it is possible to see that as the stabilising plates approach the first-wall, there is an increase in loads.



a) b)

Figure 8: The two plots (a,b) show the trend of the loads applied on the bottom plate and on the top plate for the EOF VDE-Downward FAST configuration

From the results obtained, it emerges that the loads on the top plate tend to increase as you move away from the First Wall. This behaviour can be attributed to the proximity of the plate to the Vacuum Vessel (VV) and the Eddy Currents, which circulate within it. Since the top plate is further away from the plasma, the direct interaction with the plasma becomes less significant, allowing magnetic effects due to Eddy Currents

in the VV to predominate. These currents, generated by the variation of magnetic fields, affect the load on the top plate in an increasing way as it approaches the Vacuum Vessel.

As for the bottom plate, the opposite behaviour is observed. Being closer to the plasma, the loads on it are mainly dominated by the variation in position and current of the plasma itself. The proximity amplifies the variations in the magnetic flux intercepted by the bottom plate, since in this region the flux undergoes more significant variations due to movement and current changes in the plasma. As a result, the Eddy Currents induced in the lower plate are higher than those in the upper plate, contributing predominantly to the observed loads.

Table 3: Absolute maximum values of the loads applied on the stabilizing plates, in configuration EOF VDE-Downward in the FAST case

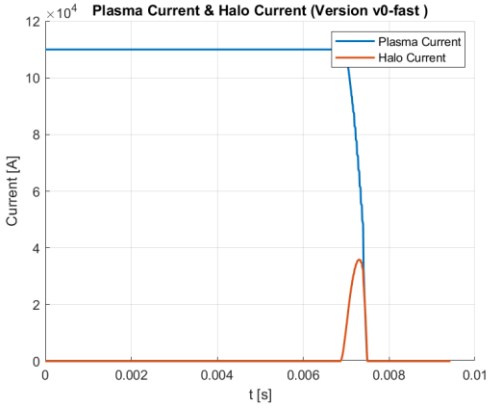
Plate Positions	|Fz| (Up) [N]	|Fz| (Bottom) [N]
V0	1379,075	2587,018
V1	1397,604	2523,824
V2	1509,923	2324,429
V3	1608,011	2053,009
V4	1687,011	1817,825
V5	1798,402	1612,465
V6	1860,307	1495,071
V7	1961,355	1428,091

This result is justified by the fact that the loads acting on the stabilising plates are generated by the "Eddy Currents", which, as shown in the following graphs, decrease as the distance of the plates from the "first-wall" increases. As a result, it could be hypothesized that it is convenient to place the stabilising plates at a greater distance from the "first-wall" to reduce electromagnetic loads.

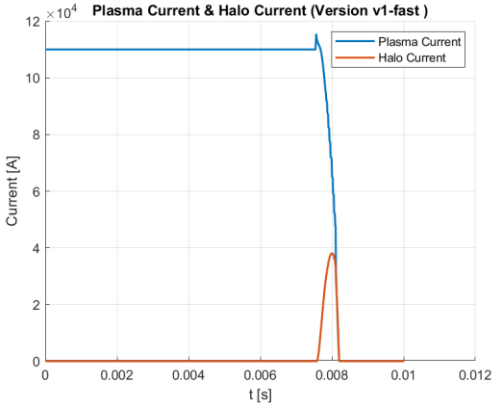
However, this choice is disadvantageous for two main reasons:

- Increased Growth Rate (GR): TRUST is a small facility and, as an experimental facility, has an intrinsically high "Growth Rate" value. A further increase in the GR, caused by a greater distance of the plates from the "first-wall", would lead to instability such as to compromise the usability of the tokamak itself.
- Halo Currents: As shown in Figure 9, the Halo Currents which, as discussed in the previous chapters, cause localized loads on the plates, tend to increase with increasing distance from the "first-wall". This is particularly relevant since the "MAXFEA" software, which is used to calculate loads, is not able to directly consider the localized contributions of Halo Currents. In fact, these currents have a distribution and MAXFEA adopts a simplified model (0D) that considers only the total intensity of the current, ignoring its spatial distribution. Therefore, the loads calculated by MAXFEA are based solely on the Eddy Currents, they do not reflect the actual contribution of the Halo Currents, which however is physically present.

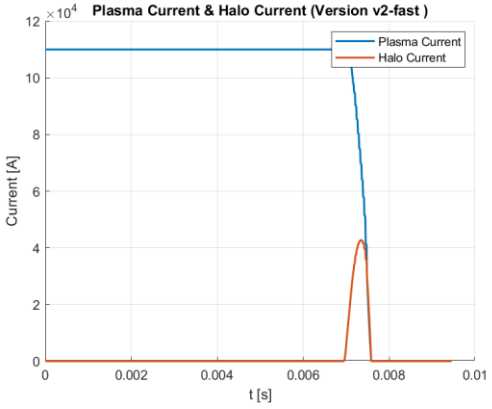
As a result, although the loads due to Eddy Currents decrease with increasing distance from the "first-wall", the Halo Currents increase. This leads to the conclusion that a configuration in which the stabilising plates are positioned close to the "first-wall", as in the case of the plate (V_0) represents the optimal choice. This position allows for an effective compromise: although the "Eddy Currents" are higher than a plate positioned further away, the "Halo Currents" are significantly lower, reducing the overall localized loads.


Plasma Current & Halo Current (Version v0-fast)
Plasma Current
Halo Current
Current [A]
t [s]

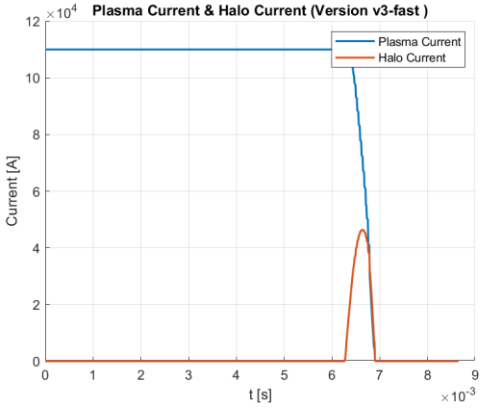
a)


Plasma Current & Halo Current (Version v1-fast)
Plasma Current
Halo Current
Current [A]
t [s]

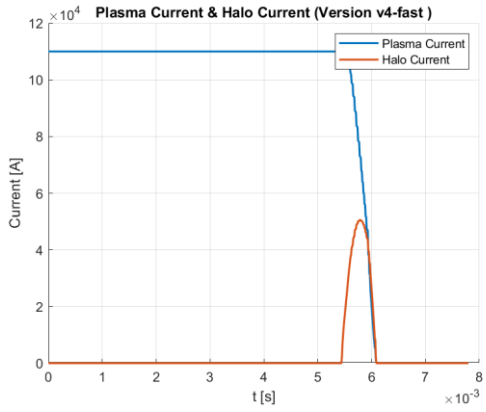
b)


Plasma Current & Halo Current (Version v2-fast)
Plasma Current
Halo Current
Current [A]
t [s]

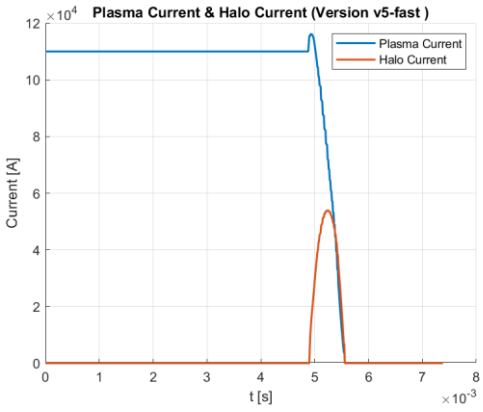
c)


Plasma Current & Halo Current (Version v3-fast)
Plasma Current
Halo Current
Current [A]
t [s]

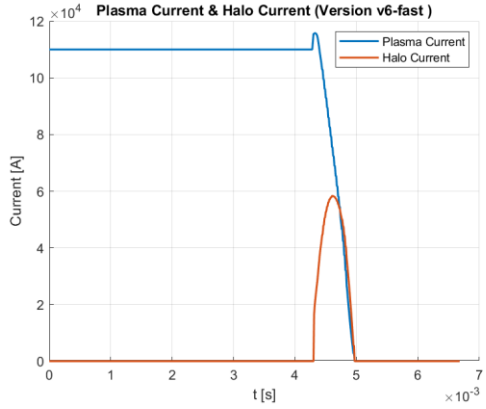
d)


Plasma Current & Halo Current (Version v4-fast)
Plasma Current
Halo Current
Current [A]
t [s]

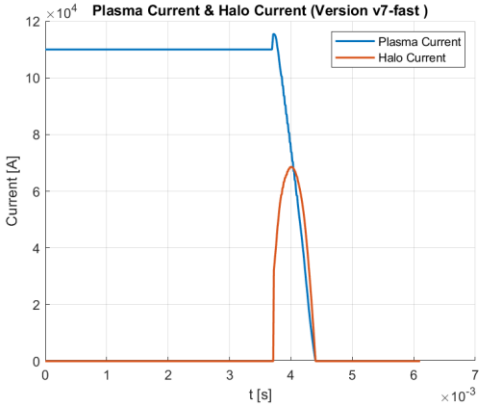
e)


Plasma Current & Halo Current (Version v5-fast)
Plasma Current
Halo Current
Current [A]
t [s]

f)



g)



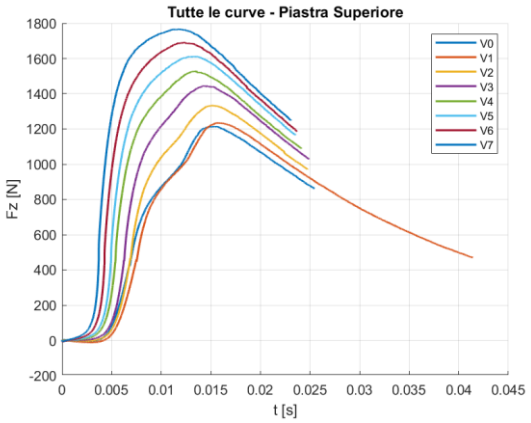
h)

Figure 9: Plots (a,b,c,d,e,f,g,h) - Trend of plasma currents and Halo Currents in the various possible positions of the plates, following an EOF VDE-Downward FAST configuration

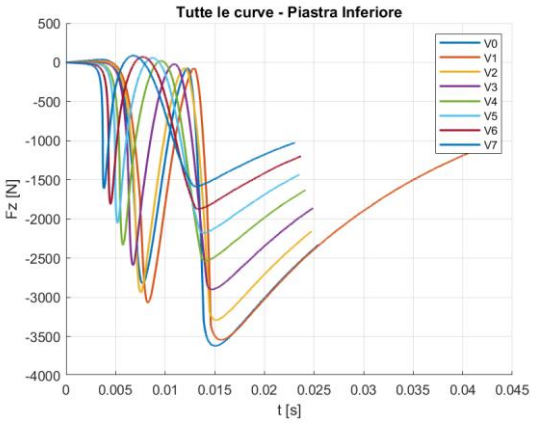
Finally, from the graphs, we can see how the "Halo Current" is reset at the exact moment when the plasma current is also cancelled, which is what we expected since the Halo Currents are a type of current that originates from the concatenation between plasma and structure, if the plasma current is zero, then the "Halo Currents" also cancel out.

3.1.3.2 Analysis Results MAXFEA VDE-Downward SLOW EOF configuration

As for the study on the EOF-VDE-Downward configuration in the SLOW case, we observe the following graphs.



a)



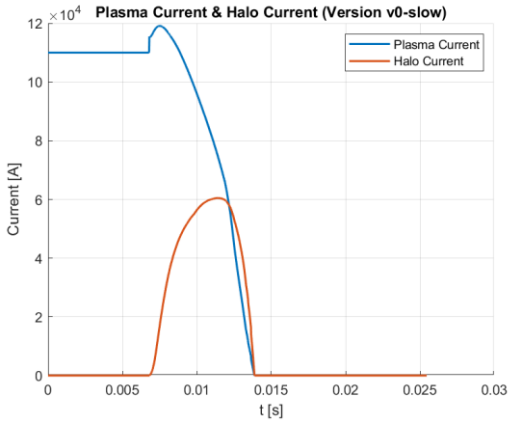
b)

Figure 10: The two plots (a,b) show the trend of the loads applied on the lower plate and on the upper plate for the EOF VDE-Downward SLOW configuration

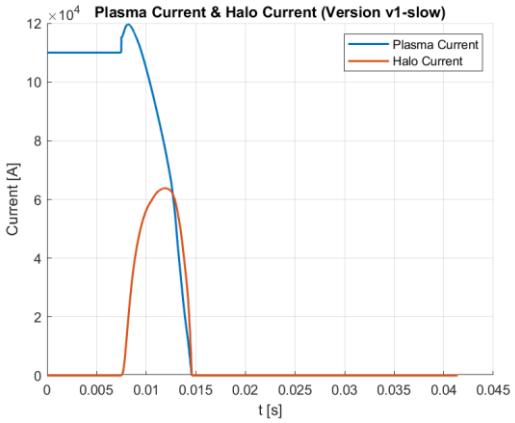
Similarly to the FAST configuration, it is possible to observe a decreasing trend of the loads applied on the bottom plate, as the distance from the first-wall increases. This is justified, as in the previous case, by the fact that the loads acting on the stabilising plates are generated by the "Eddy Currents", which decrease as the distance of the plates from the "first-wall" increases.

Table 4: Absolute maximum values of the loads applied on the stabilising plates, in VDE-Downward EOF configuration in the SLOW case

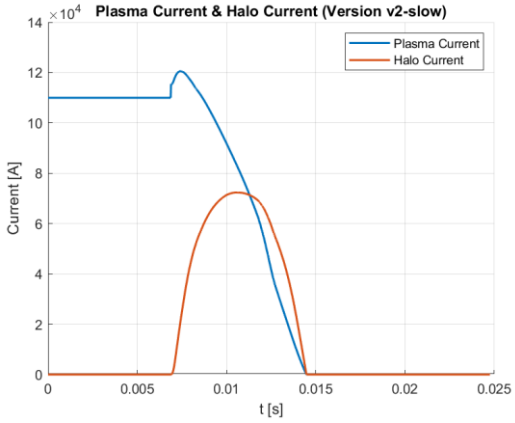
Plate Positions	|Fz| (Up) [N]	|Fz| (Bottom) [N]
V0	1214,501	3622,556
V1	1235,119	3545,602
V2	1333,533	3293,842
V3	1445,559	2902,141
V4	1529,161	2536,921
V5	1612,653	2180,988
V6	1693,039	1873,334
V7	1769,041	1611,757



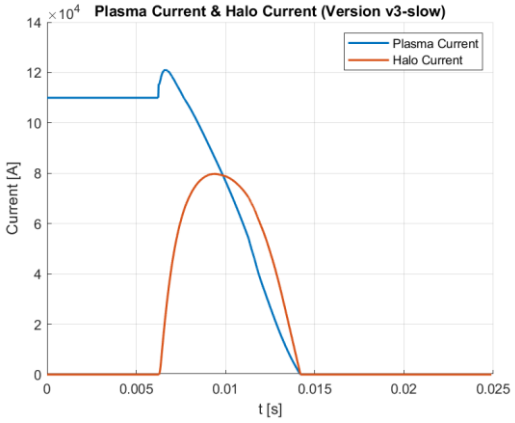
a)



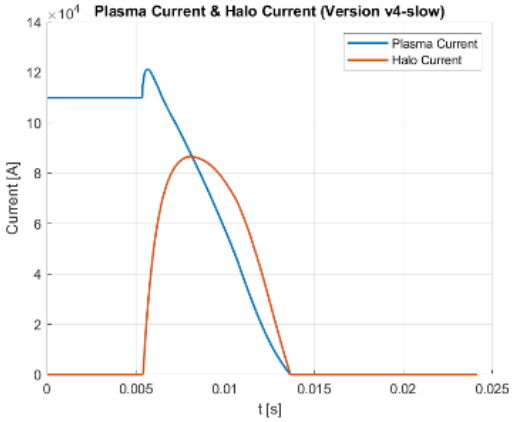
b)



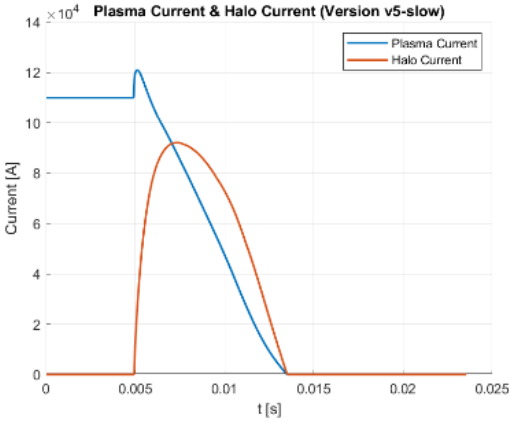
c)



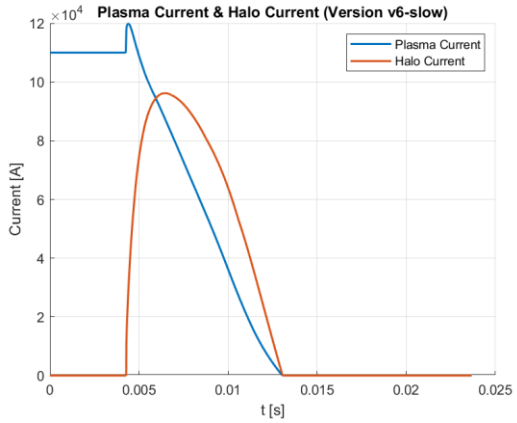
d)



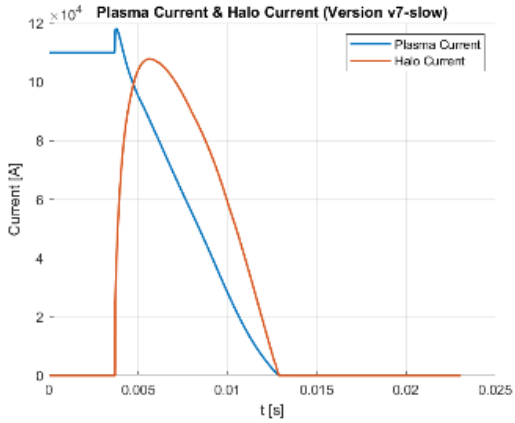
e)



f)



g)



h)

Figure 11: Plots (a,b,c,d,e,f,g,h) - Trend of plasma currents and Halo Currents in the various possible positions of the plates, following an EOF VDE-Downward SLOW configuration

We can make the same considerations as in the FAST case, regarding the Halo Currents. From the reported results, clear trends are observed for electromagnetic loads (*EM loads*) in both the FAST and SLOW cases. In both cases, it emerges that loads are generally higher when the stabilising plates are placed closer to the device's first-wall, due to the greater magnetic interaction with the confined plasma.

In the FAST case, the resulting forces on the upper plate (*Up*) vary from about **1379 N** to **1961 N**, while those on the lower plate (*Down*) are reduced from about **2587 N** to **1428 N** as the distance from the first wall increases. This indicates that, for FAST loads, the variation is more significant in the bottom plate, which is most affected by rapid plasma changes.

In the SLOW case, the loads on the upper plate start at about **1214 N** and increase to **1769 N**, while those on the lower plate decrease from **3623 N** to **1612 N** as the distance increases. Here, too, a higher intensity of loads on the bottom plate is observed, but with a more moderate trend than in the FAST case, indicating that gradual changes in the plasma have a less pronounced impact. In both cases it is possible to observe an order of magnitude equal to $10^3 N$.

4 CONCLUSIONS AND FUTURE DEVELOPMENTS

This thesis addressed the analysis of the optimization of the position of stabilising plates in the TRUST project, with the aim of improving plasma stability and mitigating electromagnetic loads generated during critical events such as disruptions. Using both the RZIP and the Filamentary models, supported by simulations with MAXFEA software, different configurations were investigated, identifying optimal positions to balance the Growth Rate (GR) and electromagnetic loads.

The results confirmed that the position of the stabilising plates significantly affects the GR. The RZIp model showed an increase in GR of up to 23% with the movement away from the first-wall compared to the selected position V_0. For the V_0 configuration, the GR value obtained is $987.7471\ s^{-1}$ with the RZIp method and $118.1019\ s^{-1}$ with the Filamentary method, which qualitatively confirmed the trend, although it was not quantitatively accurate. As far as electromagnetic loads are concerned, analysis with MAXFEA indicated that the configuration V_0 represents the best compromise between plasma stability and the loads applied to the plates. Specifically, for the lower plate, which, as explained, is the most subjected to loads, the forces are $2587.018\ N$ for the FAST case and $3622.556\ N$ for the SLOW case. This position allows minimizing the contribution of Halo Currents, which are more significant as the distance from the first-wall increases, while accepting a higher value of the global loads due to Eddy Currents. This ensures an optimal balance between global and local loads, resulting in a lower overall load than any other configuration.

On average, in the slow configuration we can observe a higher value of loads. The FAST configuration, in fact, has lower average loads than the SLOW configuration thanks to the speed with which the plasma loses electromagnetic energy in the "current quench" phase. In the FAST configuration, the derivative of the poloidal flux with respect to time is greater, generating high induced stresses, but the phenomenon is short-lived, limiting the interaction with the structures. In contrast, in the SLOW configuration, the slower descent of the current causes a greater cumulative load on the stabilising plates, as the plasma remains in contact with the structures for a longer period.

Looking ahead, the work could be extended to include three-dimensional models, advanced plate materials and simulations on different plasma configurations to evaluate in detail the loads acting on the plates. Another future development could be the study of the optimal inclination of plates. This study represents a significant contribution to improving the stability of plasma in TRUST tokamaks, providing a step forward towards optimizing the technologies needed for sustainable nuclear fusion.

The thesis contribution to the TRUST project is twofold: on the one hand, it provided an in-depth quantitative analysis of the behaviour of stabiliser plates, demonstrating the effectiveness of the proposed configuration. On the other hand, it developed a methodological approach that can be applied to other tokamak devices, improving the design and optimisation of structural components.

However, the work has some limitations: the simulations were conducted assuming an axisymmetric geometry, which may not fully capture the three-dimensional effects. In addition, the local loads due to Halo Currents have not been modelled accurately, representing an area for improvement for future research.

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